DEPARTMENT OF MECHATRONICS

MECHATRONIC SYSTEM INTEGRATION

(MCTA 3203)

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EXPERIMENT 1

DIGITAL LOGIC SYSTEM

SECTION 1

GROUP 7

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ABSTRACT

The experiment intended to design and implement a digital logic system using an Arduino CT-Uno microcontroller, a common cathode 7-segment display, two pushbuttons and ten resistors. The system was to demonstrate digital input and output interfacing by using Arduino to control the seven segments of the display to represent numerical values from 0 to 9. The circuit was assembled on a breadboard by connecting the seven display segments to individual Arduino digital pins from D2-D8 through 220Ω resistors while two pushbuttons were connected as inputs for incrementing and resetting the counter. The Arduino was programmed using the Arduino IDE and serial USB communication was used to upload the program and monitor the system through the Serial Monitor.

A simple counting algorithm was implemented in which pressing the increment button increased the displayed value sequentially from 0 to 9 while pressing the reset button returned the count back to 0. The system successfully displayed all ten numerical values and responded to the two user inputs as intended. The main technical constraints included the initial use of digital pins D0–D6 which resulted in improper display operation and required the connections to be changed to D2–D8. Moreover, other challenges encountered ensuring the correct segment wiring, selecting suitable 220Ω current-limiting resistors and achieving reliable pushbutton input detection. Potential issues such as incorrect wiring or button bouncing may affect system reliability. It is recommended to verify the proper wiring, software debouncing and additional displays or functions be implemented for future improvements.

INTRODUCTION

The objective of this experiment is to study the interfacing of a common cathode 7-segment display with a Cytron CT-UNO microcontroller. A 7-segment display consists of seven individual LED segments labelled from a to g. Different combinations of these segments are switched ON to display different numbers. By connecting the display to the microcontroller, the segments can be controlled to represent numbers from 0 to 9. In this experiment, push buttons allow the user to manually increment and reset the displayed number.

This experiment involves basic electronic circuit interfacing with digital logic systems, such as logic gates, basic ALU operations, and IC-based applications. It demonstrates how digital signals can be used to control outputs through a microcontroller. Each segment of the 7-segment display is connected through a 220Ω resistor to limit the current flow and protect the LED segments

from excessive current. The push buttons act as manual controls for the counting process. When a button is pressed, the microcontroller detects the digital input and responds by changing the number displayed on the 7-segment display. This demonstrates how user input can be used to control a digital output.

The expected outcome of this experiment is that the 7-segment display will show sequential numbers from 0 to 9 when the increment button is pressed. When the reset button is pressed, the display number will return to 0. Therefore, this experiment helps to improve the understanding of how digital inputs are processed into controlled digital outputs using a microcontroller.

MATERIALS AND EQUIPMENTS

The experiment was conducted using the following components :

- Cytron CT-UNO
- Common cathode 7-segment display
- 220Ω resistors (7 units)
- 10kΩ resistors (2 units)
- Pushbuttons (2 units)
- Jumper wires
- Breadboard
- Arduino IDE

EXPERIMENTAL SETUP AND HARDWARE ASSEMBLY

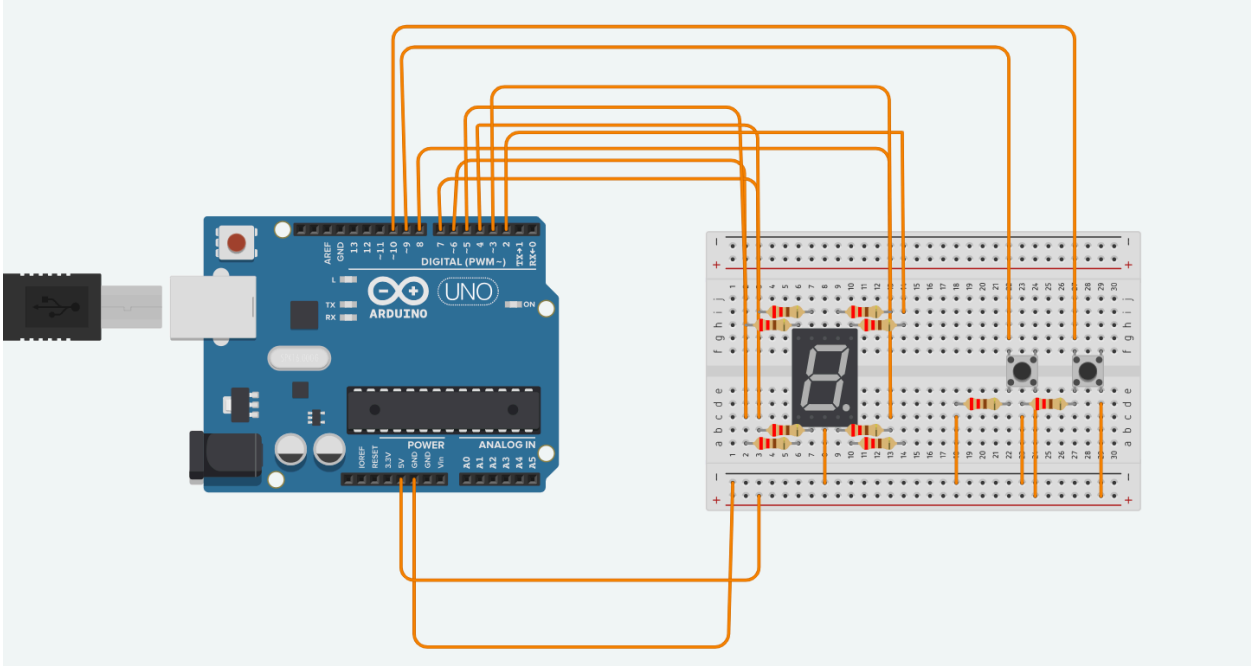


Figure shows the schematic diagram for the 7-segments display

1. The common cathode 7-segment display was connected to the Cytron CT-UNO with each of the seven segments labelled A to G connected to a separate digital pin of the Arduino from D2 to D8.
2. The common cathode pin of the display was connected to one of the GND (ground) pins on the Arduino.
3. Each segment pin was connected to its corresponding Arduino digital pin through a 220Ω resistor to limit the current.
4. The push buttons were connected to the Arduino with one leg of each push button connected to a separate digital pin of D9 and D10 while the other leg was connected to GND.
5. A 10-kΩ pull-up resistor was connected to each push button with one end connected to the corresponding digital pin and the other end connected to the 5V output of the Arduino.

METHODOLOGY AND PROGRAM LOGIC

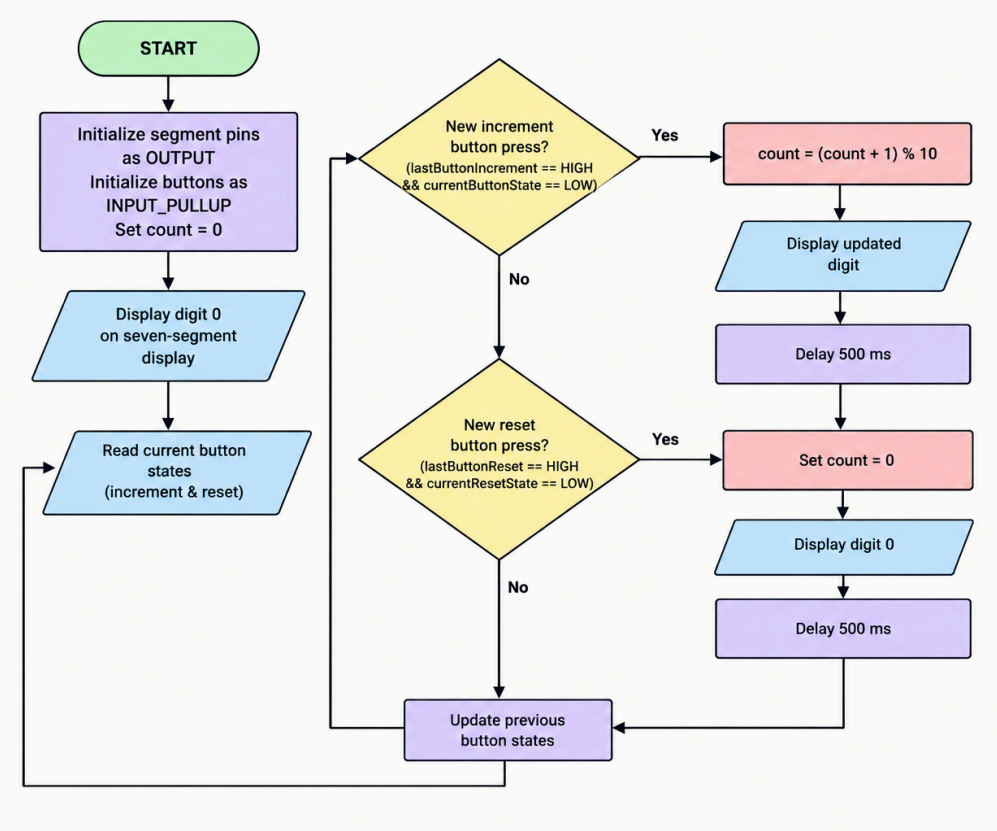


Figure shows flowchart of 7-Segments Display Counter with Increment and Reset Functions

Experiment steps:

1. The circuit was built according to the circuit setup instructions.
2. Upload the provided IDE code into Arduino Uno.
3. The increment button was pressed to increase the count. The 7-segment display should show the number starting from 0 until 9 and back to 0 after 9.
4. The reset button was pressed to reset the count to 0.

IDE Code:

```
const int segmentA = 2;
const int segmentB = 3;
const int segmentC = 4;
const int segmentD = 5;
const int segmentE = 6;
const int segmentF = 7;
```

```
const int segmentG = 8;
const int buttonIncrement = 9;
const int buttonReset = 10;

int count = 0;
bool lastButtonIncrement = HIGH;
bool lastButtonReset = HIGH;

void displayDigit(int num){
  switch (num){
    case 0:
      digitalWrite(segmentA, HIGH);
      digitalWrite(segmentB, HIGH);
      digitalWrite(segmentC, HIGH);
      digitalWrite(segmentD, LOW);
      digitalWrite(segmentE, HIGH);
      digitalWrite(segmentF, HIGH);
      digitalWrite(segmentG, HIGH);
      break;

    case 1:
      digitalWrite(segmentA, HIGH);
      digitalWrite(segmentB, LOW);
      digitalWrite(segmentC, LOW);
      digitalWrite(segmentD, LOW);
      digitalWrite(segmentE, LOW);
      digitalWrite(segmentF, LOW);
      digitalWrite(segmentG, HIGH);
      break;

    case 2:
      digitalWrite(segmentA, HIGH);
      digitalWrite(segmentB, HIGH);
```

```
digitalWrite(segmentC, LOW);
digitalWrite(segmentD, HIGH);
digitalWrite(segmentE, HIGH);
digitalWrite(segmentF, HIGH);
digitalWrite(segmentG, LOW);
break;

case 3:
  digitalWrite(segmentA, HIGH);
  digitalWrite(segmentB, HIGH);
  digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, HIGH);
  digitalWrite(segmentE, LOW);
  digitalWrite(segmentF, HIGH);
  digitalWrite(segmentG, HIGH);
  break;

case 4:
  digitalWrite(segmentA, HIGH);
  digitalWrite(segmentB, LOW);
  digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, HIGH);
  digitalWrite(segmentE, LOW);
  digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, HIGH);
  break;

case 5:
  digitalWrite(segmentA, LOW);
  digitalWrite(segmentB, HIGH);
  digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, HIGH);
  digitalWrite(segmentE, LOW);
```

```
  digitalWrite(segmentF, HIGH);
  digitalWrite(segmentG, HIGH);
  break;

case 6:
  digitalWrite(segmentA, LOW);
  digitalWrite(segmentB, HIGH);
  digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, HIGH);
  digitalWrite(segmentE, HIGH);
  digitalWrite(segmentF, HIGH);
  digitalWrite(segmentG, HIGH);
  break;

case 7:
  digitalWrite(segmentA, HIGH);
  digitalWrite(segmentB, HIGH);
  digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, LOW);
  digitalWrite(segmentE, LOW);
  digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, HIGH);
  break;

case 8:
  digitalWrite(segmentA, HIGH);
  digitalWrite(segmentB, HIGH);
  digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, HIGH);
  digitalWrite(segmentE, HIGH);
  digitalWrite(segmentF, HIGH);
  digitalWrite(segmentG, HIGH);
  break;
```

```
    case 9:
      digitalWrite(segmentA, HIGH);
      digitalWrite(segmentB, HIGH);
      digitalWrite(segmentC, HIGH);
      digitalWrite(segmentD, HIGH);
      digitalWrite(segmentE, LOW);
      digitalWrite(segmentF, HIGH);
      digitalWrite(segmentG, HIGH);
      break;
  }
}

void setup() {
  pinMode(segmentA, OUTPUT);
  pinMode(segmentB, OUTPUT);
  pinMode(segmentC, OUTPUT);
  pinMode(segmentD, OUTPUT);
  pinMode(segmentE, OUTPUT);
  pinMode(segmentF, OUTPUT);
  pinMode(segmentG, OUTPUT);
  pinMode(buttonIncrement, INPUT_PULLUP);
  pinMode(buttonReset, INPUT_PULLUP);
  displayDigit(count);
}

void loop() {
  bool currentButtonState = digitalRead(buttonIncrement);
  bool currentResetState = digitalRead(buttonReset);

  if (lastButtonIncrement == HIGH && currentButtonState == LOW) {
    count = (count + 1) % 10;
    displayDigit(count);
```

```
    delay(500);
  }

  if (lastButtonReset == HIGH && currentResetState == LOW){
    count = 0;
    displayDigit(count);
    delay(500);
  }

  lastButtonIncrement = currentButtonState;
  lastButtonReset = currentResetState;
}
```

DATA COLLECTION AND TECHNICAL ANALYSIS

The data was collected by observing the number displayed on the common cathode 7-segment display when the push buttons were pressed. The increment button was pressed repeatedly to observe the counting sequence from 0 to 9, while reset button was pressed to show that the displayed number returned to 0. The observed results are presented in the tables below.

Button Pressed	Previous Number	Displayed Number
Initial	-	0
Increment (1)	0	1
Increment (2)	1	2
Increment (3)	2	3
Increment (4)	3	4
Increment (5)	4	5
Increment (6)	5	6
Increment (7)	6	7
Increment (8)	7	8
Increment (9)	8	9

Table shows an incrementing sequence of the 7-segment display

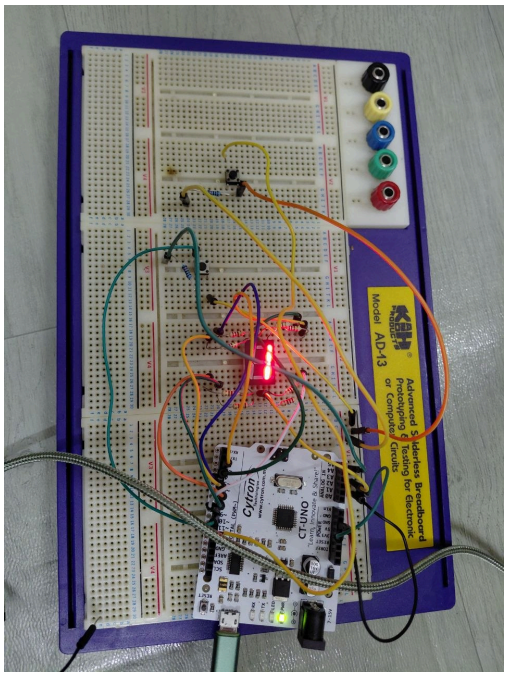
Button Pressed	Previous Number	Displayed Number
Initial	-	0
Reset	6	0

Table shows the reset function of the 7-segment display

Digit	Segment Active	Observed Display
0	A,B,C,E,F,G	Correct
1	A,G	Correct
2	A,B,D,E,F	Correct
3	A,B,D,F,G	Correct
4	A,C,D,G	Correct
5	B,C,D,F,G	Correct
6	B,C,D,E,F,G	Correct
7	A,B,G	Correct
8	A,B,C,D,E,F,G	Correct
9	A,B,C,D,F,G	Correct

Table shows the segment activation for digits 0-9 on 7-segment displays

RESULT AND ENGINEERING DISCUSSION



Question

How can you interface an I2C LCD with Arduino? Explain the coding principle behind it compared to a 7-segment display and a matrix LED.

Answer

An I2C LCD with Arduino can be interfaced by using only two pins which are SDA and SCL pin through the I2C protocol where it allows Arduino to send data with simple commands with libraries like LiquidCrystal_I2C. This will make the coding easier and reduce wiring. As we can see, a 7-segment display requires multiple digital pins to control each segment and numbers formed by turning segments ON and OFF manually. LED matrix uses the concept of row-column scanning which requires complex coding and timing control to display patterns. Overall, I2C LCD is the simplest and most efficient while the 7-segment display and matrix LED require more control and complexity.

CONCLUSION

The experiment successfully achieved its objective of interfacing a 7-segment display with Arduino CT-Uno and controlling it using one push button displaying increment by 1 from 0 to 9 each time the button pressed and the other button to reset the counter. It enhanced understanding of digital logic systems, hardware interfacing, and basic programming for embedded applications.

RECOMMENDATIONS

Identify each segment before starting the experiment by using codes to identify which segment is which since not every 7-segment display contains the same segment position from A to G which may cause the display to not match the desired number or pattern. Software debounce also can be implemented to improve button reliability and prevent false triggers. The system can be expanded to display multiple digits. Applying bit-shifting by using IC can also reduce Arduino pin usage.

REFERENCES

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https://youtu.be/_JbHhfZvfGU?si=SHa6H_f7JVFJ2Bxu

ACKNOWLEDGEMENT

We would like to express our deepest appreciation to our instructor and lab personnel for their guidance and support throughout this experiment.

STUDENT'S DECLARATION

We hereby declare that the work presented in this technical report is entirely our own, except where explicitly acknowledged. We certify that in developing and completing this mechatronics project, we have strictly adhered to the standards of academic integrity and have not engaged in any form of plagiarism or unethical conduct. All sources of information, reference materials, and technical support utilized in this work have been appropriately cited and acknowledged.

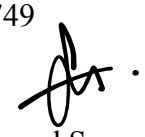
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This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been undertaken or done by unspecified sources or persons. We hereby certify that this report has not been done by only one individual, and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have read and understand the content of the total report, and no further improvement on the reports is needed from any of the individual

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